COMPUTER NETWORKS
Module 3 - NETWORK LAYER

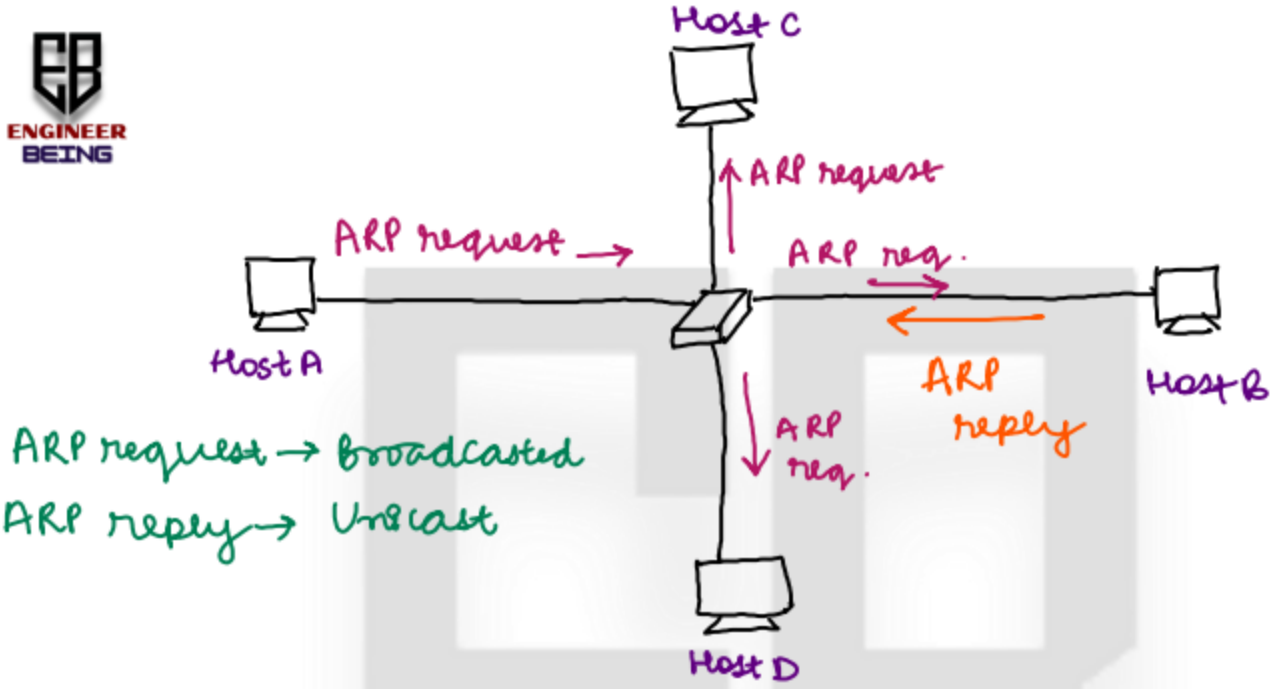
Network Layer: Point-to-point networks, Logical addressing, Basic internetworking (IP, CIDR, ARP, RARP, DHCP, ICMP), Routing, forwarding and delivery, Static and dynamic routing, Routing algorithms and protocols, Congestion control algorithms, IPv6.

Divide the network with IP address 200.1.2.0 into 5 subnets. 10M
Discuss the role of logical addressing. 2M
Discuss about the IP ranges of Class A, B, C and D. 2M
Write advantages of Next-generation IPV6 over IPV4. 2M
The IP network 200.198.160.0 is using subnet mask 255.255.255.224. Design the subnets. 10M
Discuss in detail about ICMP role in network layer. 10M
What is unicast routing? Discuss unicast routing protocols. 10M
What is the difference between network layer delivery and transport layer delivery? 2M
If a class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet? 10M
What is the count-to-infinity problem? 2M

ADDRESS MAPPING

Mapping Logical to Physical Address: ARP

Every host or router on the network receives and processes the ARP query packet, but only the intended recipient recognizes its IP address and sends back an ARP response packet. The response packet contains the recipient's IP and physical addresses. The packet is unicast directly to the inquirer by using the physical address received in the query packet



ARP request → Broadcasted

ARP reply → Unicast

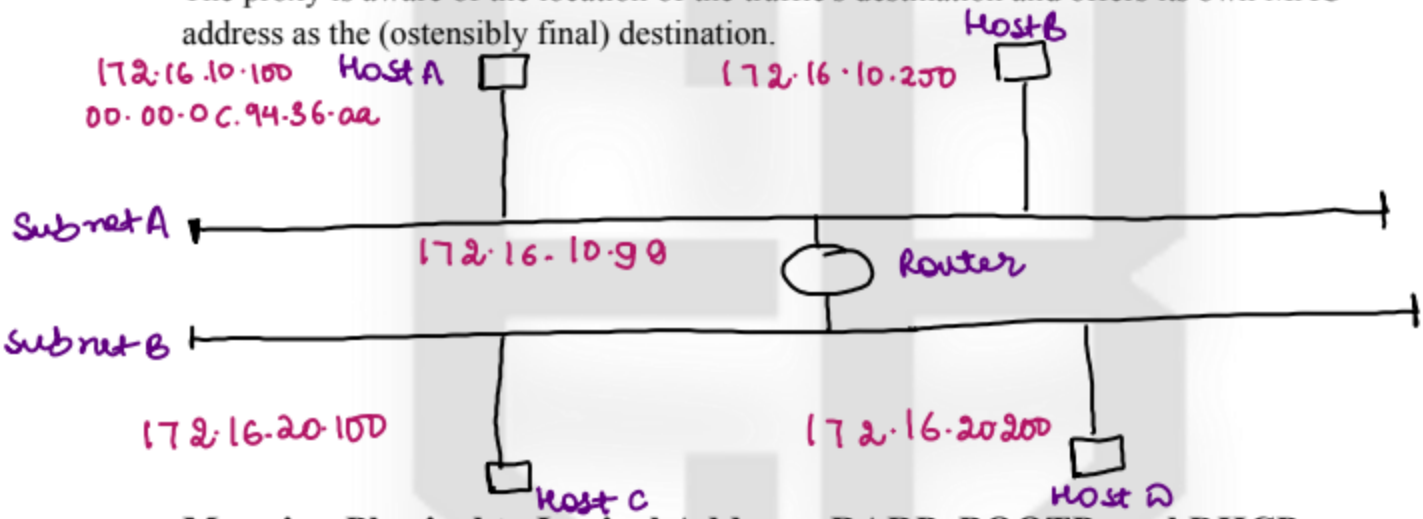
Hardware type		Protocol type
Hardware Length	Protocol Length	Operation Request 1, Reply 2
Sender hardware address (6 bytes)		
Sender Protocol Address (4 Bytes IP)		
Target Hardware Address (6 Byte)		
Target Protocol Address (4 Byte)		

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Proxy ARP

Proxy ARP is a technique by which a proxy server on a given network answers the Address Resolution Protocol (ARP) queries for an IP address that is not on that network. The proxy is aware of the location of the traffic's destination and offers its own MAC address as the (ostensibly final) destination.



Mapping Physical to Logical Address: RARP, BOOTP, and DHCP

RARP

A RARP request is created and broadcast on the local network.

Another machine on the local network that knows all the IP addresses will respond with a RARP reply.

The requesting machine must be running a RARP client program; the responding machine must be running a RARP server program.

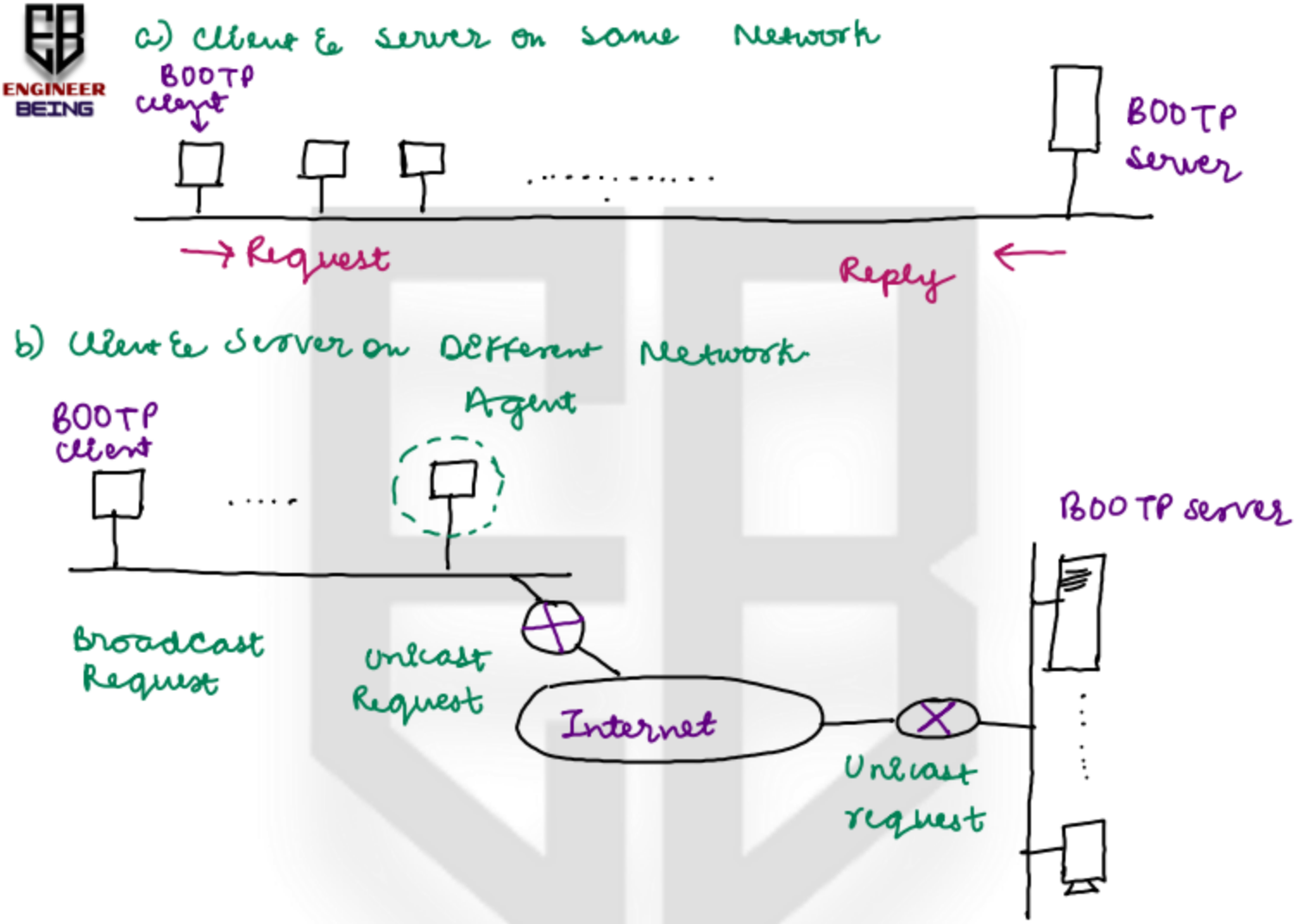
Drawback

There is a serious problem with RARP: Broadcasting is done at the data link layer. The physical broadcast address, allis in the case of Ethernet, does not pass the boundaries of a network.

This means that if an administrator has several networks or several subnets, it needs to assign a RARP server for each network or subnet. This is the reason that RARP is almost obsolete. 1\vo protocols, BOOTP and DHCp, are replacing RARP.

BOOTP

The Bootstrap Protocol (BOOTP) is a client/server protocol designed to provide physical address to logical address mapping. The administrator may put the client and the server on the same network or on different networks.



In this scenario, the BOOTP request's broadcast IP datagram can't pass through the router. To resolve this, a vendor extension is used as an intermediary to act as the relay agent.

The relay agent is configured with the unicast address. When the agent receives the broadcast packet, it sends the unicast diagram to the BOOTP server. This is then routed to the BOOTP client.

BOOTP is not a dynamic configuration protocol. When a client requests its IP address, the BOOTP server consults a table that matches the physical address of the client with its IP address. This implies that the binding between the physical address and the IP address of the client already exists. The binding is predetermined. However, what if a host moves from one physical network to another? What if a host wants a temporary IP address?

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BOOTP cannot handle these situations because the binding between the physical and IP addresses is static and fixed in a table until changed by the administrator. BOOTP is a static configuration protocol.

DHCP

DHCP provides static and dynamic address allocation that can be manual or automatic.

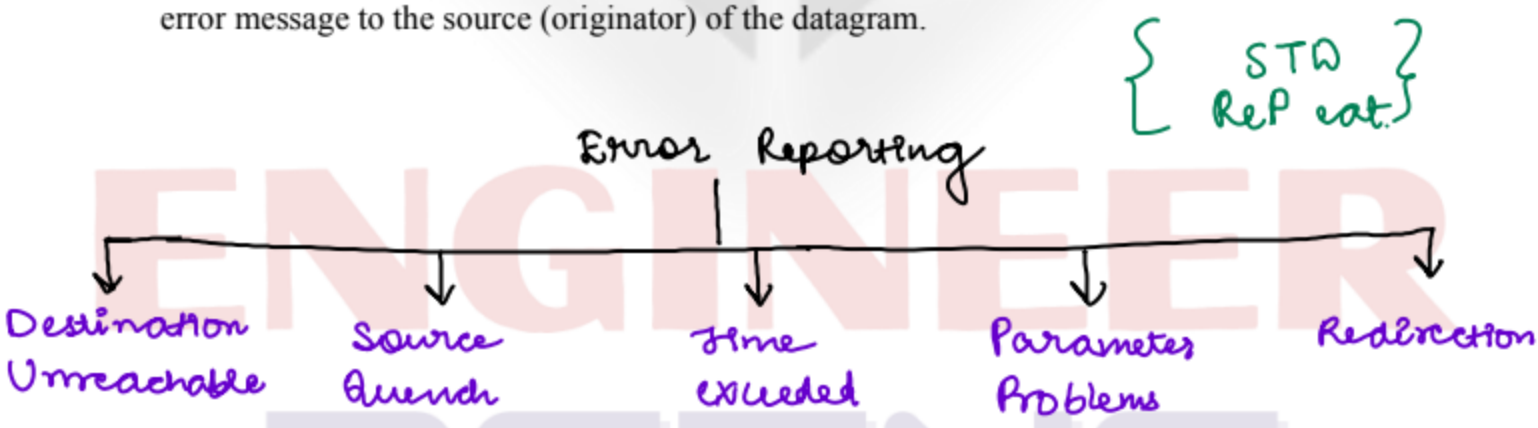
Static Address Allocation: In this capacity DHCP acts as BOOTP does.A DHCP server has a database that statically binds physical addresses to IP addresses

Dynamic Address Allocation DHCP has a second database with a pool of available IP addresses. This second database makes DHCP dynamic. When a DHCP client requests a temporary IP address, the DHCP server goes to the pool of available (unused) IP addresses and assigns an IP address for a negotiable period of time.

ICMP(Internet Control Message Protocol)

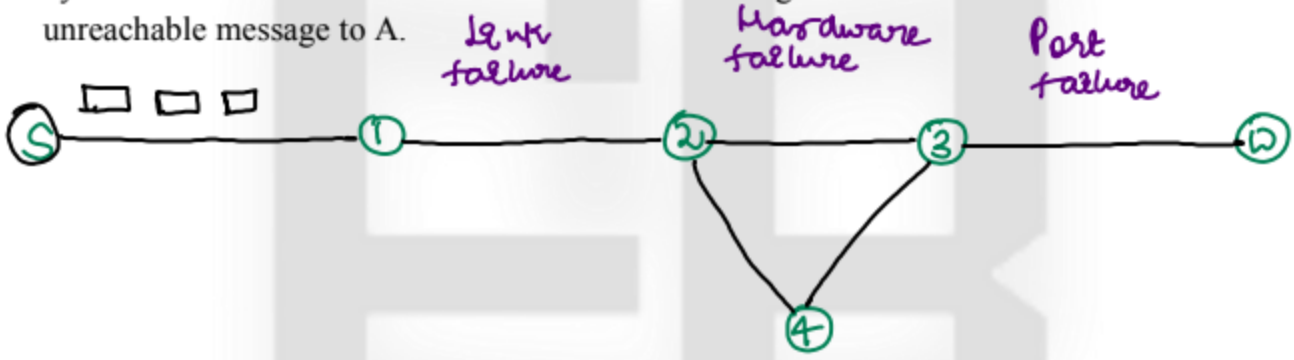
ICMP messages are divided into two broad categories: error-reporting messages and query messages.

The error-reporting messages report problems that a router or a host (destination) may encounter when it processes an IP packet. ICMP uses the source IP address to send the error message to the source (originator) of the datagram.



Destination Unreachable

The host/router sends this message if it is not able to route the IP packet to its destination. For example, if sender A wants to send the datagram to receiver B but it is not received by B then the intermediate router will discard the datagram and send the destination unreachable message to A.



Source Quench

- If the datagrams are received much faster than they can be forwarded or processed, the queue may overflow. In this case, the router or the host has no choice but to discard some of the datagrams. The source-quench message in ICMP was designed to add a kind of flow control to the IP.
- When a router or host discards a datagram due to congestion, it sends a source-quench message to the sender of the datagram.

This message has two purposes:

- First, it informs the source that the datagram has been discarded.
- Second, it warns the source that there is congestion somewhere in the path and that the source should slow down (quench) the sending process.

Time Exceeded

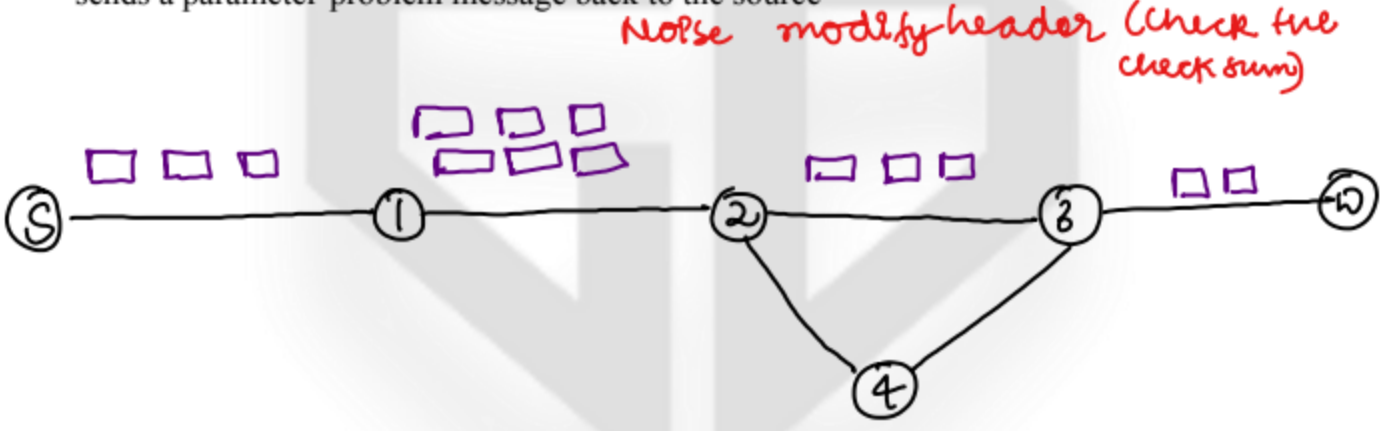
When a datagram visits a router, the value of this field is decremented by 1. When the time-to-live value reaches 0, after decrementing, the router discards the datagram. However, when the datagram is discarded, a time-exceeded message must be sent by the router to the original source. Second, a time-exceeded message is also generated when not all fragments that make up a message arrive at the destination host within a certain time limit.



ICMP Time Exceeded

Parameter Problem

Any ambiguity in the header part of a datagram can Create serious problems as the datagram travels through the Internet. If a router or the destination host discovers an ambiguous or missing value in any field of the datagram, it discards the datagram and sends a parameter-problem message back to the source



Redirection Concept

The host/router sends this message to update the routing table of the host.
For example, sender A wants to send the message to receiver B and there is a router between them. Then, A sends the data to the router and the router sends the message to B and redirection message to A so that A can update its routing table.

Query Messages

- Echo Request & Reply
- Timestamp Request & Reply
- Address-mask Request & Reply
- Router Solicitation & Advertisement

Echo Request and Reply

The combination of echo-request and echo-reply messages determines whether two systems (hosts or routers) can communicate with each other.
ping command that can create a series (instead of just one) of echo-request and echo-reply messages.
For example, host A wants to check whether it can communicate with another host B so it will send an echo request to B and if there is a link between A and B and B is active then it will send an echo reply to A on receiving the request.

Timestamp Request and Response

The ICMP Timestamp Request and Timestamp Reply query messages provide the ability to determine the length of time that ICMP query messages spend in transit, which is extremely useful for measuring the latency across a specific network.

Address-Mask Request and Reply

- A host may know its IP address, but it may not know the corresponding mask.
- To obtain its mask, a host sends an address-mask-request message to a router on the LAN.
- If The host knows the address of the router, it sends the request directly to the router. If it does not know, it broadcasts the message.
- The router receiving the address-mask-request message responds with an address-mask-reply message, providing the necessary mask for the host

For example, a host may know its IP address as 159.31.17.24, but it may not know that the corresponding mask is /24.

Router Solicitation and advertisement

It is used to confirm the presence of a router on the local area network. It is done by broadcasting a router solicitation message and the router that receives the message broadcasts the routing information using a router advertisement message.

For example, if host A wants to get the information of routers present on the network. It will broadcast a router solicitation message to request routers to advertise their presence and in response routers on the network will announce its IP address available for routing through an advertisement message.

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IPV4

•An IPv4 address is a 32-bit address that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet.

•The IPv4 addresses are unique and universal.

Notations

•There are two prevalent notations to show an IPv4 address

Binary Notation

•01110101 10010101 00011101 00000010

Dotted-Decimal Notation

•117.149.29.2



Ques. Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Ques. Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 221.34.7.82

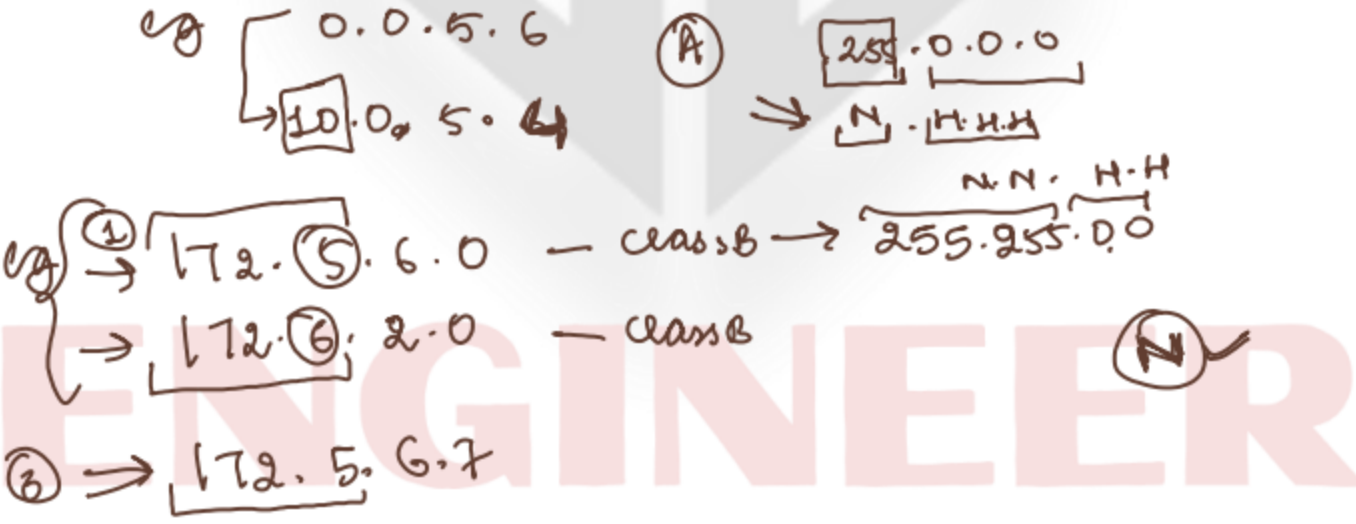
Classful Addressing

In classful addressing, the address space is divided into five classes:

A, B, C, D, and E.

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Class	1st octet range in Decimal	1st octet bits	Network and host portion	Subnet Mask	Number of possible host and networks
A	0-127	0	N.H.H.H	255.0.0.0	128 Nets (2^7) 16,777,214 hosts (2^{24}-2)
B	127-191	10	N.N.H.H	255.255.0.0	16,284 Nets (2^{14}) 65,534 hosts (2^{16}-2)
C	192-223	110	N.N.N.H	255.255.255.0	2,09,150 Nets(2^{21}) 254 hosts (2^8-2)
D	224-239	1110	multicast	-	-
E	240-255	1111	experimental	-	-



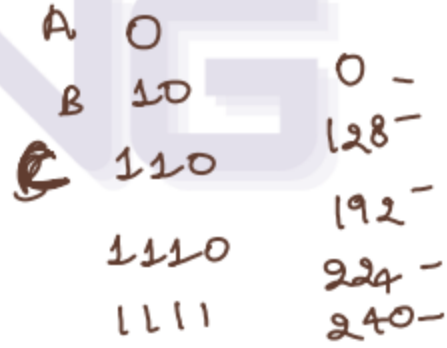
Ques. Find the class at each address.

a. 0000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

c. 14.23.120.8

d. 252.5.15.111



Subnet Mask:

A subnet mask is used to divide an IP address into two parts. One part identifies the host (computer), the other part identifies the network to which it belongs.

N.H.H.H
255.0.0.0

Ex: 10.10.10.1 — A
10.10.20.16 — A

Ex: 10.10.10.1 A
10.10.20.16 A

N.N.H.H
255.255.00

Ex: 172.16.200.1 — B
172.16.165.2 — B

⇒ 172.15.165.2

Ques. A host assigned an IP address 192.168.17.9. Find the number of addresses in the block, the first and last address.

I. find class - C

II. write subnet mask - N.N.N.(H)B ↓ 255.255.255.0

11111111

0 - 127
128 - 191
192 -
224 -
240 -

Network = 192.168.17.0 - 192.168.17.255

no of addresses = $2^8 - 2$

usable addresses =

first address = 192.168.17.0

last Address = 192.168.17.255

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Ques. A host has been assigned an IP address 185.28.17.9. Find the number of addresses in the block, the first and last address.

(B)

Class B

Subnet mask: N.N.H.H → 255.255.0.0 ⇒ 16

Network: 185.28.0.0 – 185.28.255.255

no of addresses – 2^{16}

Usable address – $2^{16} - 2$

1st address – 185.28.0.0

last address – 185.28.255.255

Subnet mask – 11111111.11111111.11110000.11110000

CLASSLESS ADDRESSING (Classless Inter Domain Routing)

In this scheme, there are no classes, but the addresses are still granted in blocks.

Ques. A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address in the block?

Ans. 205.16.37.32.

Operation: AND

205.16.37.39 — 11001101.00010000.00100101.00100111

255.255.255.240 — 11111111.11111111.11111111.11110000

Ques. Find the last address for the block.

Ans 205.16.37.47

Operation: OR

205.16.37.32 — 11001101.00010000.00100101.00100000

1's Complement 255.255.255.240 — 00000000.00000000.00000000.00001111

PRIVATE IP ADDRESSES

Host that do not require access to the internet can use private addresses

10.0.0.0 to 10.255.255.255

172.16.0.0 to 172.31.255.255

192.168.0.0 to 192.168.255.255

SUBNETTING

Subnetting should be done in such a way that the network does not get affected. This means that we can divide the network into different parts but all when put together should perform the same task when done before splitting into small parts.

Ques. Subnet the IP address 150.15.0.0 into 500 hosts in each subnet.

I. Class

II. Subnet Mask -

In Binary

III no. of host = Subnet Generator = Octet

IV New Subnet mask.

V Network Range (Subnets)

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Divide the network with IP address 200.1.2.0 into 5 subnets. 10M

I. Class

II Subnet Mask- in Binary

III. no. of subnets

PREVIOUS YEAR ★

1.Discuss about the IP ranges of Class A, B, C and D. 2M
2.What is the subnetwork number of a host with an IP address of 172.16.66.0/21? 2M
3.If a class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet? 2M
4.The IP network 200.198.160.0 is using subnet mask 255.255.255.224. Design the subnets. 10M

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0
127
192
224
240

No. of addresses, Last & first address, No. of hosts, Useful addresses

Ques. Divide the network with IP address 200.1.2.0 into 5 hosts in each subnet.

0 - 127
128 - 191
→ 192 - 223
224 - 239
240 - 255.

I. class C N N N . H

II. Subnet mask- 255.255.255.0
1-1.1-1.1-1.00000000

III. host = 5 = 101 = 3 bits SQ = 8 octet = 4

1-1.1-1.1-1.1111 1 000 (8 4 2 1) ★ 128+64+32+16+8

new subnet mask- 255.255.255.248

IV. Range

200.1.2.0 - 200.1.2.7
200.1.2.8 - 200.1.2.15
16 - 1 - 31
32 -
40 -
48 -

Ques. Divide the network with IP address 200.1.2.0 into 5 subnets. 10M

I. class C

II. subnet mask 255.255.255.0
1-1.1-1.1-1.00000000

III. subnets = 5 = 101 = 3 bits SQ = 32 octet = 4
128+64+32

new subnet mask- 1-1.1-1.1-1.111 00000
255.255.255.224

IV Range

200.1.2.0 - 200.1.2.31
200.1.2.32 - 200.1.2.63
200.1.2.64 - 200.1.2.95
96
128 - 127

Ques. Subnet the IP address 150.15.0.0 into 500 hosts in each subnet.

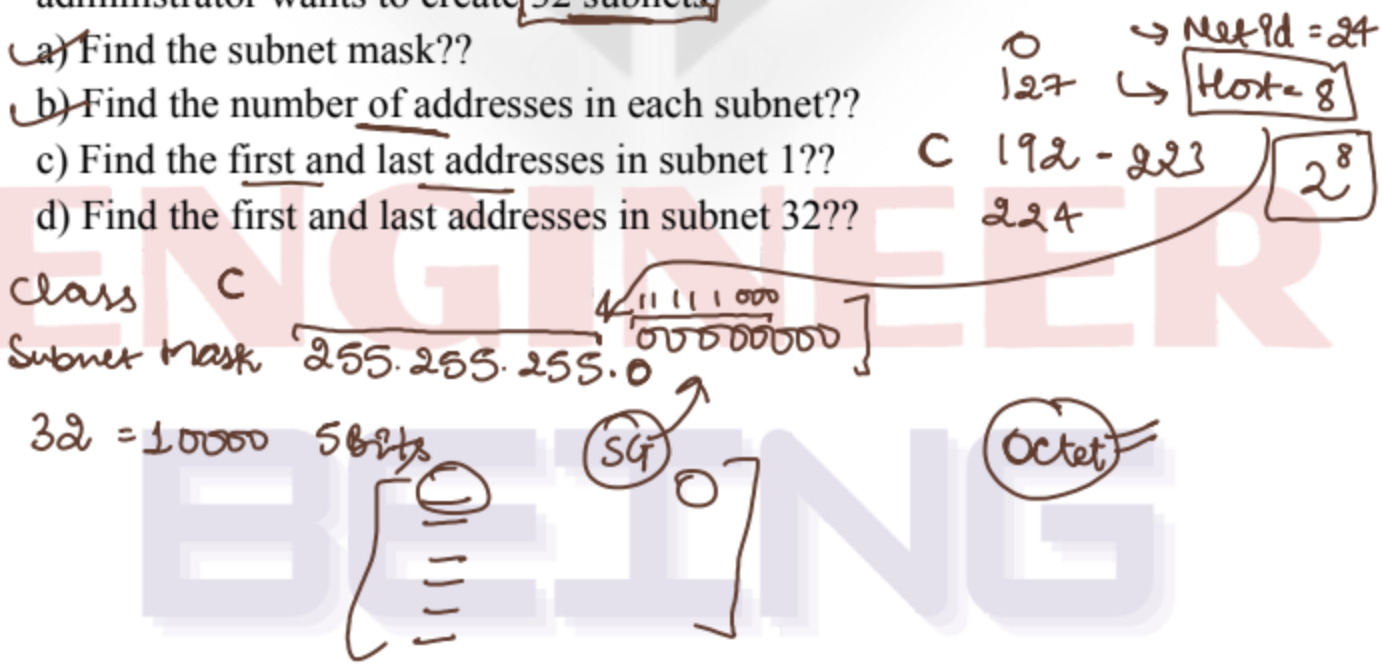
Ques: An organization is granted the block 211.17.180.0/24. The administrator wants to create 32 subnets.

a) Find the subnet mask??

b) Find the number of addresses in each subnet??

c) Find the first and last addresses in subnet 1??

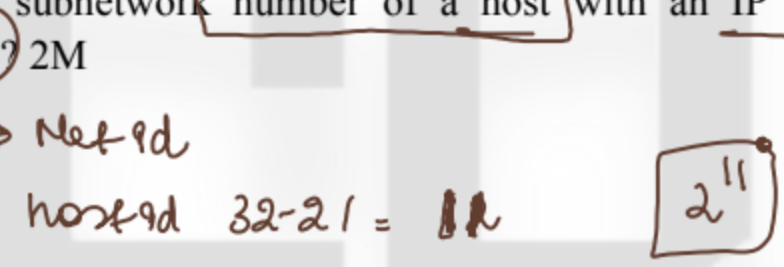
d) Find the first and last addresses in subnet 32??



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PREVIOUS YEAR QUESTIONS

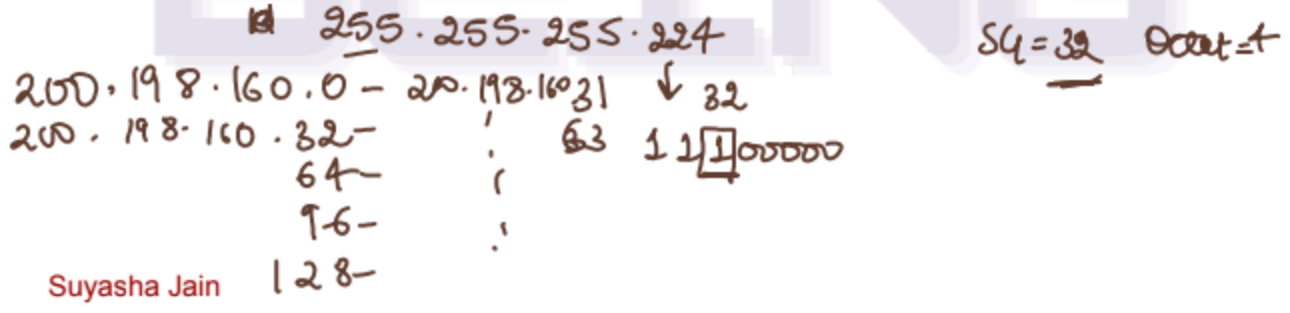
1. Discuss about the IP ranges of Class A, B, C and D. 2M
2. What is the subnetwork number of a host with an IP address of 172.16.66.0/21? 2M



3. If a class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet? 2M



4. The IP network 200.198.160.0 is using subnet mask 255.255.255.224. Design the subnets



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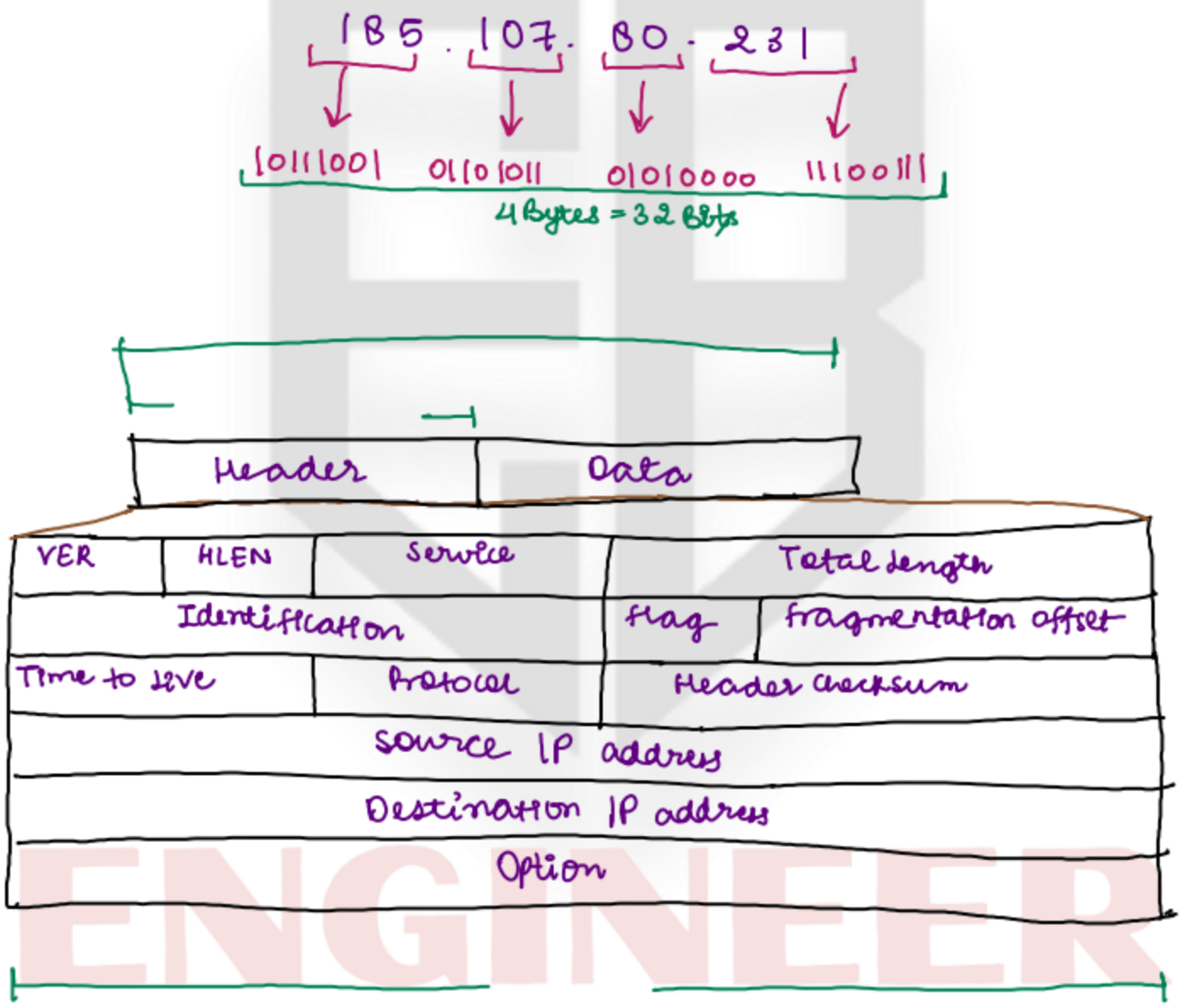
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IPV4 Datagram



Ques. In an IPv4 packet, the value of HLEN is 1000 in binary. How many bytes of options are being carried by this packet?

Ans: The HLEN value is 8, which means the total number of bytes in the header is 8 x 4, or 32 bytes. The first 20 bytes are the base header, the next 12 bytes are the options.

IPV6

Also known as IPng (Internetworking Protocol, next generation also known as IPng (Internetworking Protocol, next generation)

An IPv6 address consists of **16 bytes (octets); it is 128 bits long**

Ques. Expand the address 0:15::1:12:1213 to its original.

Ex- FDEC : 0074 : 0000 : 0000 : 0000 : B0FF : 0000 : FFFF

FDEC

Advantages of IPV6 over IPV4

•**Larger address space** An IPv6 address is 128 bits long. Compared with the 32-bit address of IPv4, this is a huge (2^96) increase in the address space.

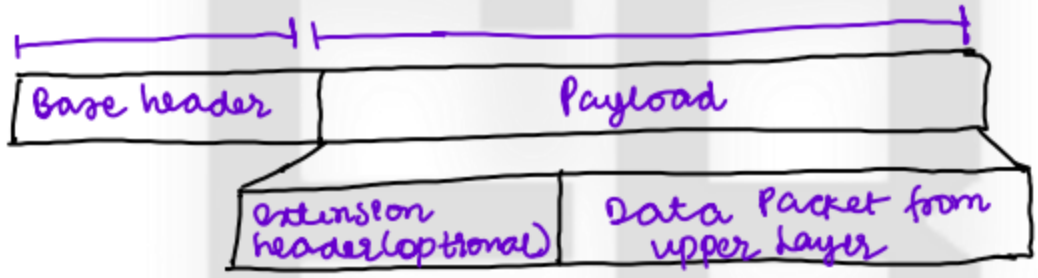
•**Better header format** IPv6 uses a new header format in which options are separated from the base header and inserted, when needed, between the base header and the upper-layer data. .

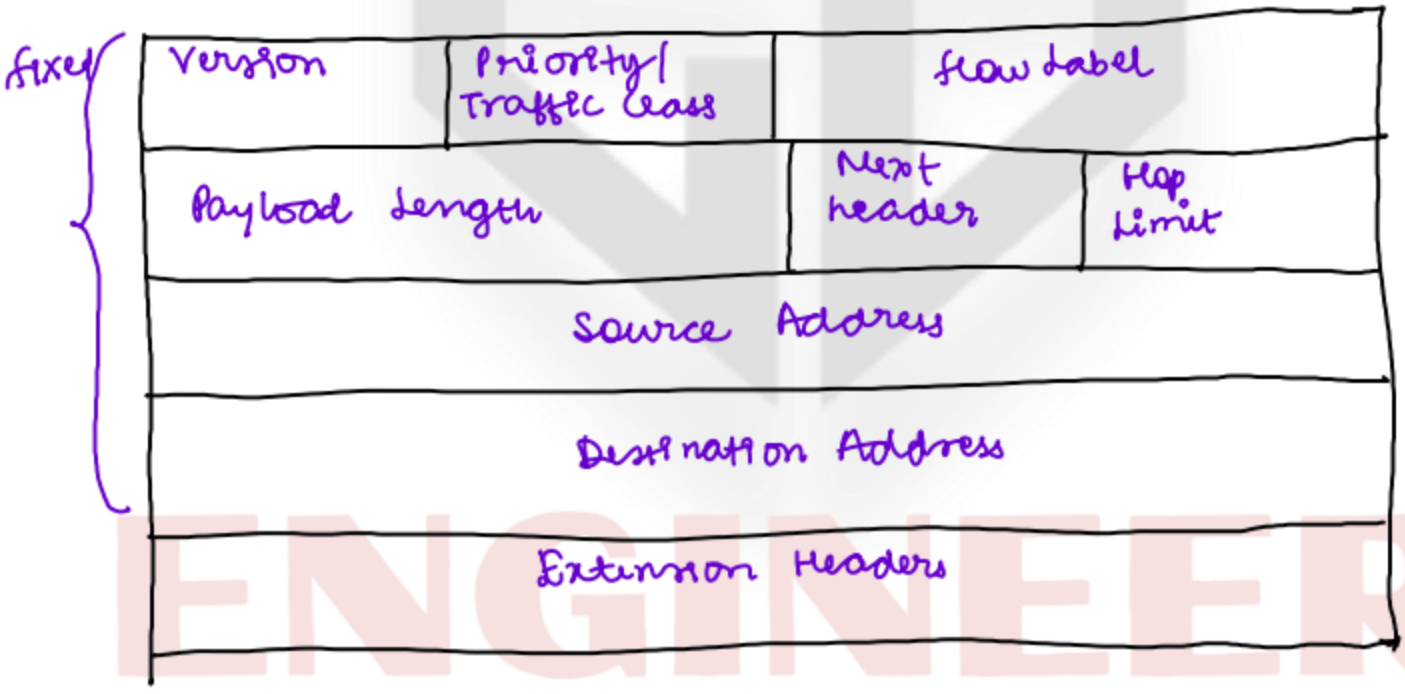
•**New options**. IPv6 has new options to allow for additional functionalities.

•**Allowance for extension** IPv6 is designed to allow the extension of the protocol if required by new technologies or applications.

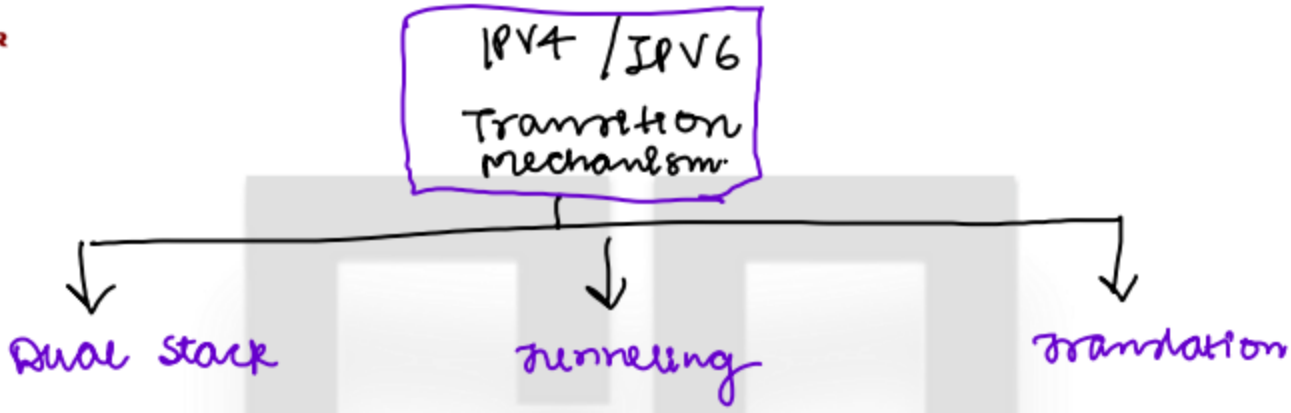
•**Support for resource allocation**

•**Support for more security**





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Dual Stack

•It is recommended that all hosts, before migrating completely to version 6, have a dual stack of protocols.

•In other words, a station must run IPv4 and IPv6 simultaneously until all the Internet uses IPv6.

Tunneling

•Tunneling is a strategy used when two computers using IPv6 want to communicate with each other and the packet must pass through a region that uses IPv4.

•To pass through this region, the packet must have an IPv4 address.

•So the IPv6 packet is encapsulated in an IPv4 packet when it enters the region, and it leaves its capsule when it exits the region

Header Translation

•Header translation is necessary when the majority of the Internet has moved to IPv6 but some systems still use IPv4.

•The sender wants to use IPv6, but the receiver does not understand IPv6.

•Tunneling does not work in this situation because the packet must be in the IPv4 format to be understood by the receiver.

•In this case, the header format must be totally changed through header translation

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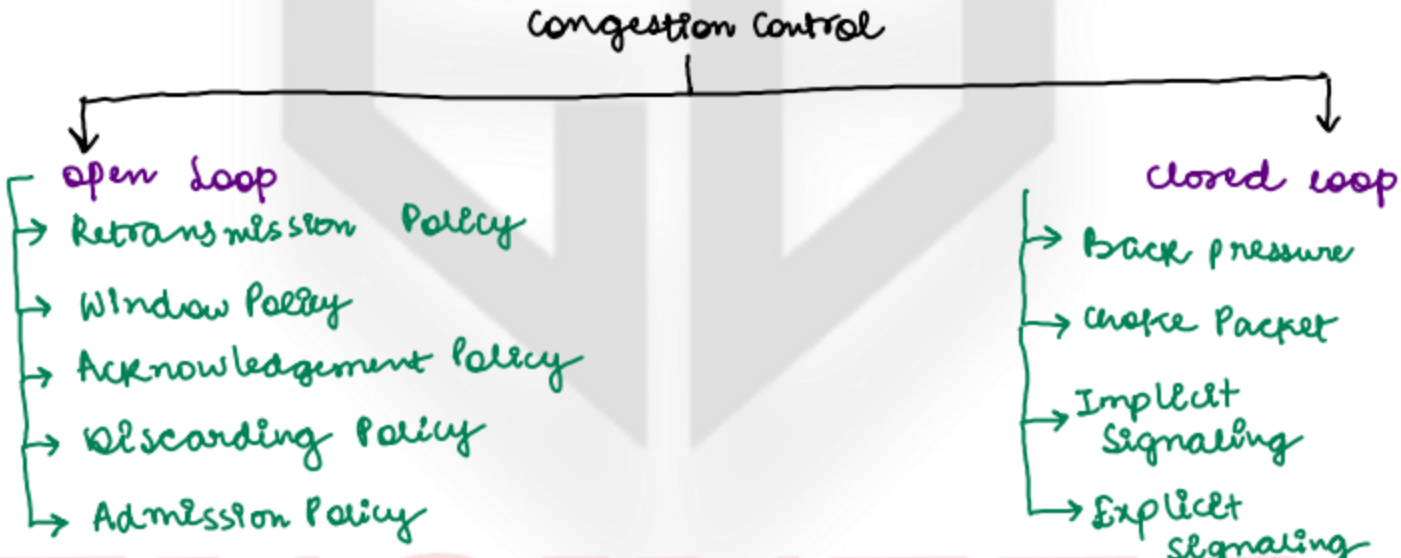
Routing Protocols in Network Layer

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CONGESTION CONTROL AND QUALITY OF SERVICE

- Congestion in a network may occur if the load on the network-the number of packets sent to the network-is greater than the capacity of the network-the number of packets a network can handle.
- Congestion control refers to the mechanisms and techniques to control congestion and keep the load below the capacity.
- An important issue in a packet-switched network is congestion.

we can divide congestion control mechanisms into two broad categories: open-loop congestion control (prevention) and closed-loop congestion control (removal)



Open-Loop Congestion Control
In open-loop congestion control, policies are applied to prevent congestion before it happens. In these mechanisms, congestion control is handled by either the source or the destination.

Retransmission Policy

- If the sender feels that a sent packet is lost or corrupted, the packet needs to be retransmitted.
- Retransmission in general may increase congestion in the network. However, a good retransmission policy can prevent congestion.
- The retransmission policy and the retransmission timers must be designed to optimize efficiency and at the same time prevent congestion.

Window Policy

- The type of window at the sender may also affect congestion.
- The Selective Repeat window is better than the Go-Back-N window for congestion control.
- In the Go-Back-N window, when the timer for a packet times out, several packets may be resent, although some may have arrived safe and sound at the receiver. This duplication may make the congestion worse.
- The Selective Repeat window, on the other hand, tries to send the specific packets that have been lost or corrupted.

Acknowledgement Policy

- If The receiver does not acknowledge every packet it receives, it may slow down the sender and help prevent congestion.
- A receiver may send an acknowledgment only if it has a packet to be sent or a special timer expires.
- A receiver may decide to acknowledge only N packets at a time.

Discarding Policy

A discarding policy by the routers prevents congestion and at the same time may not harm the integrity of the transmission.

Admission Policy

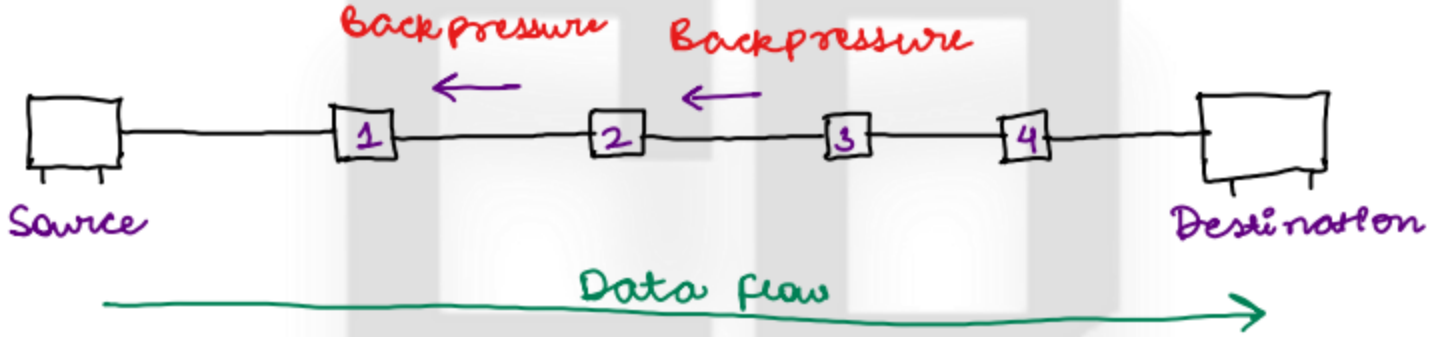
An admission policy, which is a quality-of-service mechanism, can also prevent congestion in virtual-circuit networks. Switches in a flow first check the resource requirement of a flow before admitting it to the network. A router can deny establishing a virtual circuit connection if there is congestion in the network or if there is a possibility of future congestion.

CLOSED LOOP CONGESTION CONTROL

1. Backpressure:

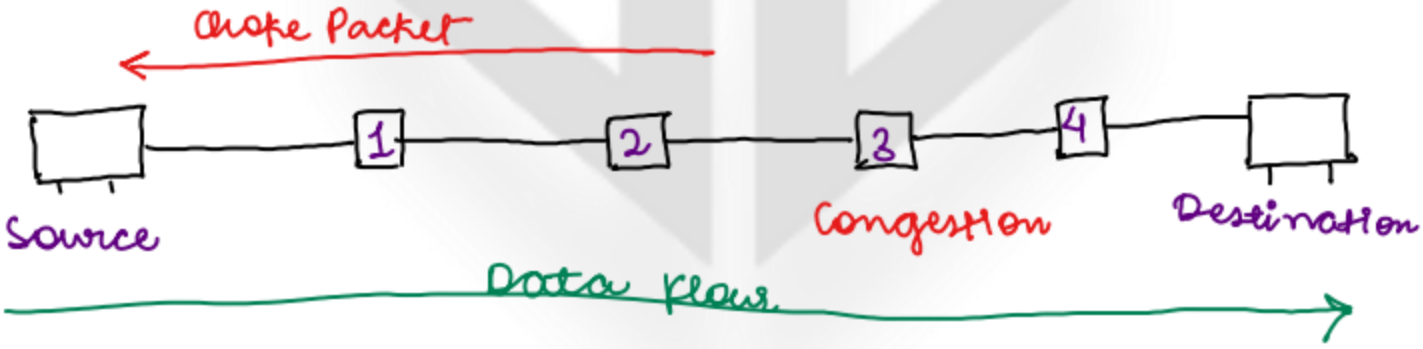
Backpressure is a technique in which a congested node stops receiving packets from upstream nodes. This may cause the upstream node or nodes to become congested and reject receiving data from above nodes. Backpressure is a node-to-node congestion control technique that propagates in the opposite direction of data flow. The backpressure

technique can be applied only to virtual circuits where each node has information of its above upstream node.



2. Choke Packet Technique :

Choke packet technique is applicable to both virtual networks as well as datagram subnets. A choke packet is a packet sent by a node to the source to inform it of congestion.



3. Implicit Signaling :

In implicit signaling, there is no communication between the congested nodes and the source. The source guesses that there is congestion in a network. For example when the sender sends several packets and there is no acknowledgement for a while, one assumption is that there is congestion.

4.Explicit Signaling :

In explicit signaling, if a node experiences congestion it can explicitly sends a packet to the source or destination to inform about congestion. The difference between choke packet and explicit signaling is that the signal is included in the packets that carry data rather than creating a different packet as in case of choke packet technique.

Explicit signaling can occur in either forward or backward direction.

Forward Signaling : In forward signaling, a signal is sent in the direction of the congestion. The destination is warned about congestion. The receiver in this case adopt policies to prevent further congestion.

Backward Signaling : In backward signaling, a signal is sent in the opposite direction of the congestion. The source is warned about congestion and it needs to slow down.

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IPV4

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Routing Protocols in Network Layer

•Routing refers to the way routing tables are created to help in forwarding.

•Forwarding means to place the packet in its route to its destination.

•Forwarding requires a host or a router to have a routing table.

•A host or a router has a routing table with an entry for each destination, or a combination of destinations, to route packets.

•The routing table can be either static or dynamic.

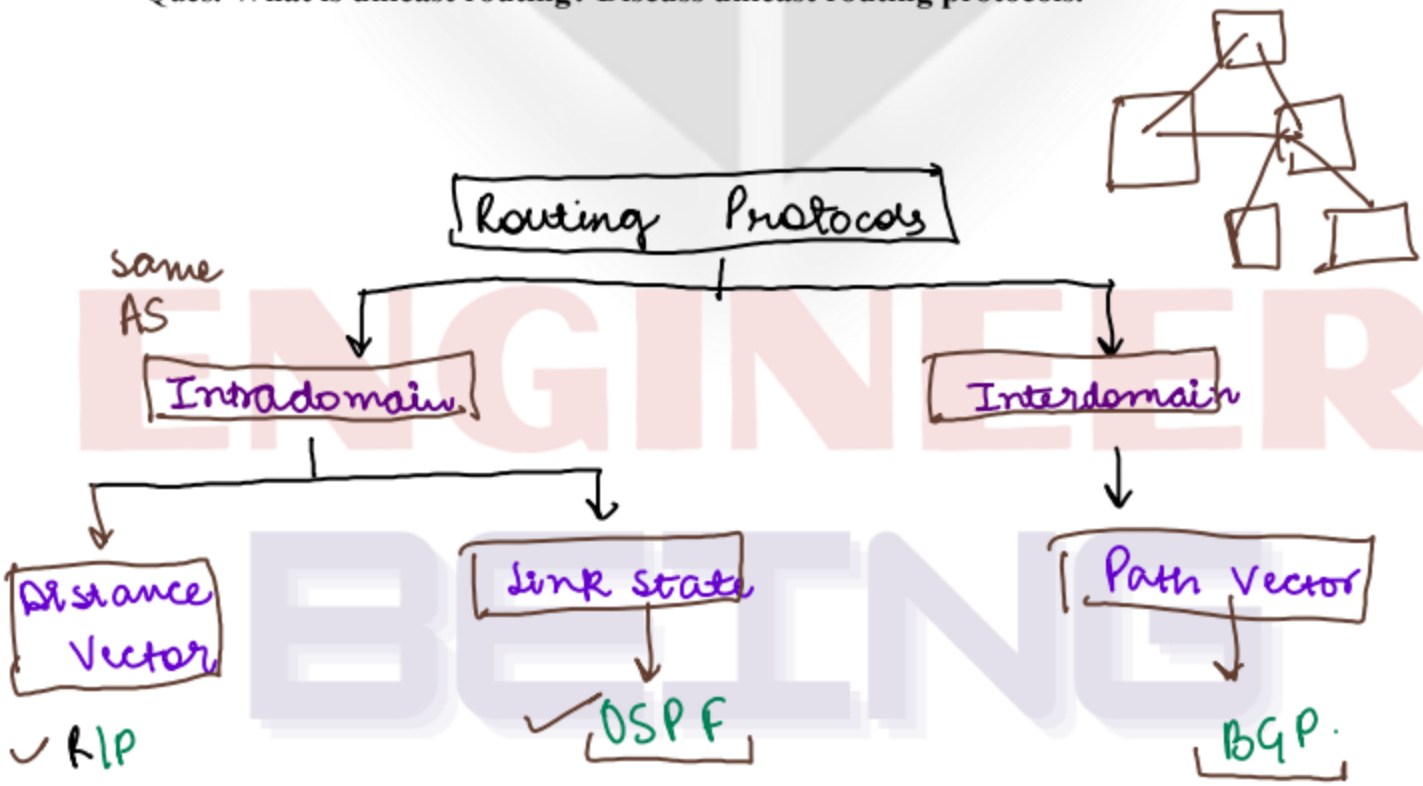
•A routing table can be either static or dynamic.

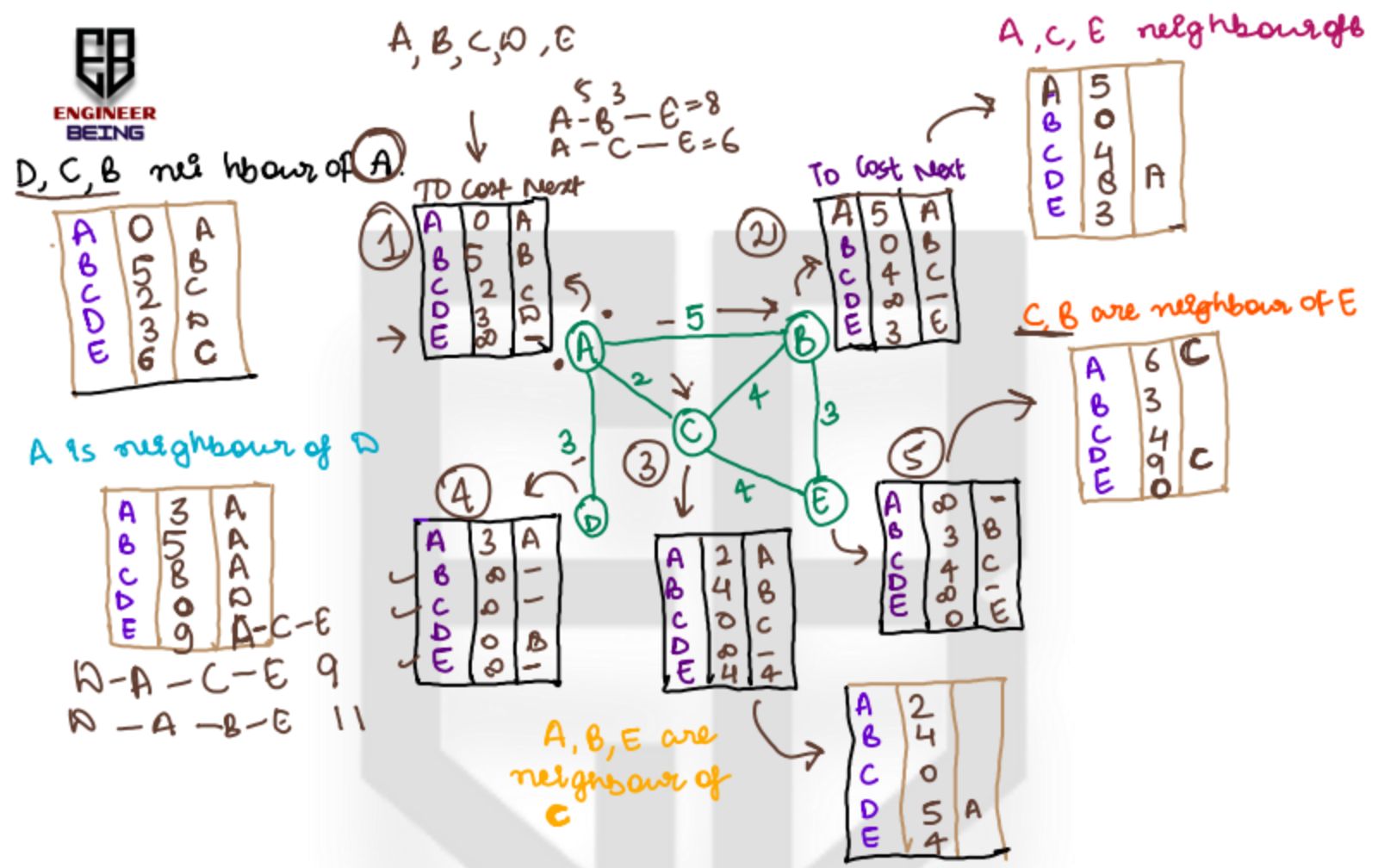
• A static table is one with manual entries.

•A dynamic table, on the other hand, is one that is updated automatically when there is a change somewhere in the internet.

Ques. **Explain distance vector routing (DVR) with working example in detail.**

Ques. **What is unicast routing? Discuss unicast routing protocols.**





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RIP

•The Routing Information Protocol (RIP) is an intradomain routing protocol used inside an autonomous system. It is a very simple protocol based on distance vector routing.

Disadvantage:

Count-to-infinity problem: The count to infinity problem happens in routing when routers get confused about how to reach a network after a problem. They keep increasing the number of steps it takes to reach it, causing a loop. This can lead to slow and incorrect routing. Techniques like split horizon help solve this problem.

Link State Routing

2.

•In link state routing, if each node in the domain has the entire topology of the domain the list of nodes and links

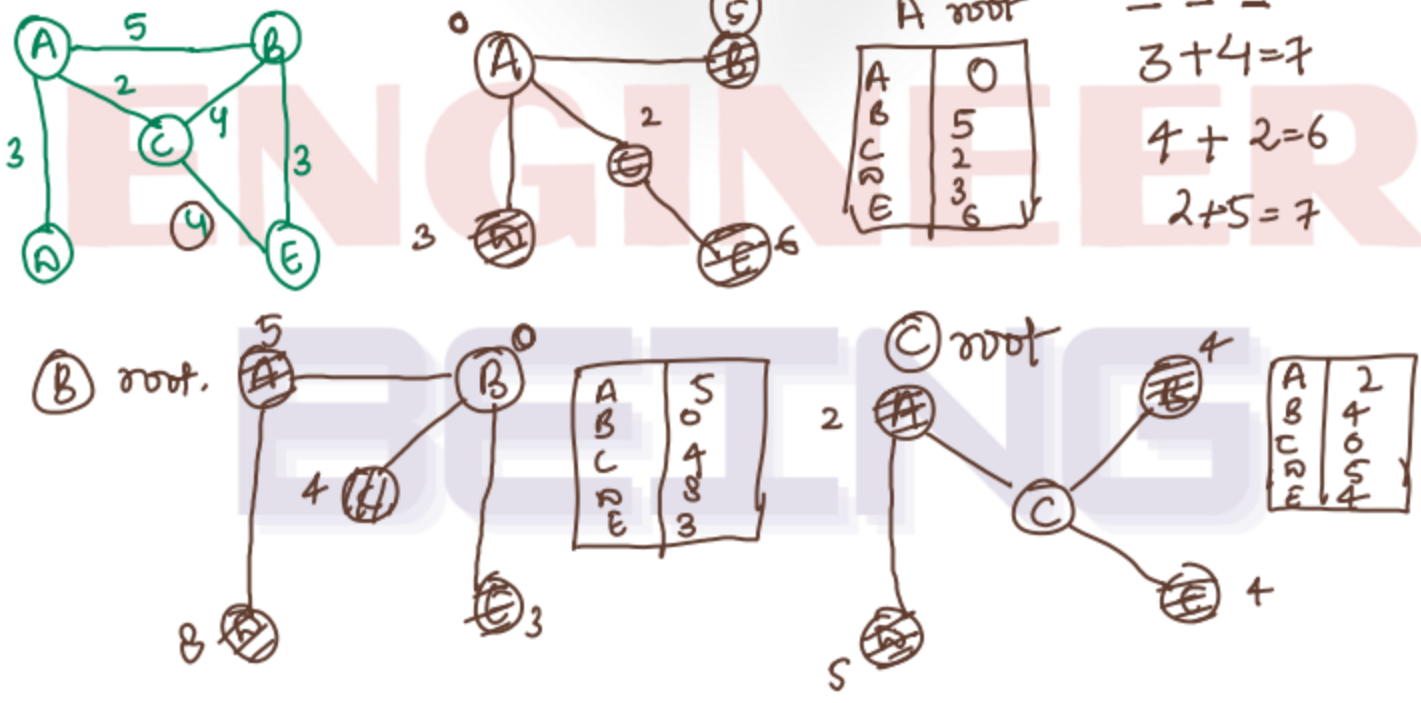
•The node can use Dijkstra's algorithm to build a routing table.

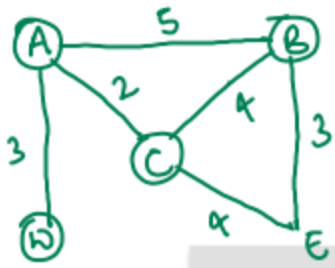
•Calculation of a routing table based on the shortest path tree.

OSPF

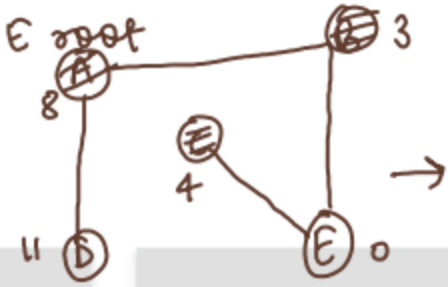
•The Open Shortest Path First or OSPF protocol is an intradomain routing protocol based on link state routing.

•Its domain is also an autonomous system.



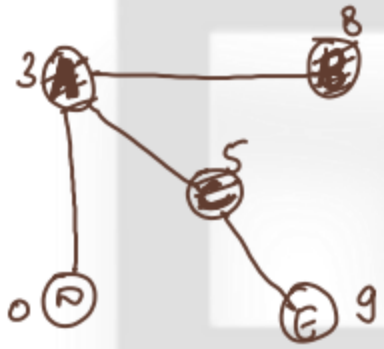


(x) E root



A	8
B	3
C	4
D	11
E	0

(*) root



A	3
B	8
C	5
D	0
E	9

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Path Vector Routing

•Distance vector and link state routing are both intradomain routing protocols.

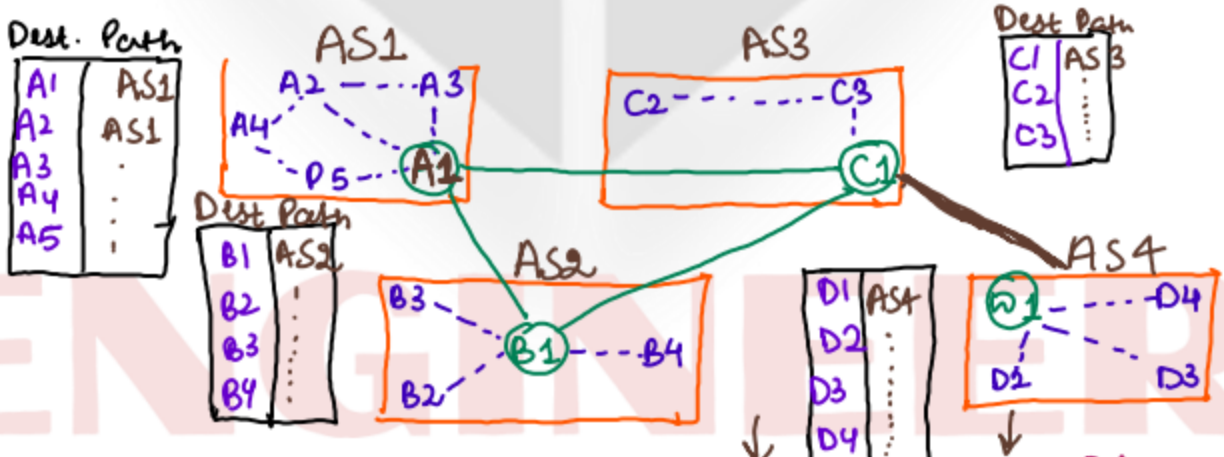
•Path vector routing proved to be useful for interdomain routing. The principle of path vector routing is similar to that of distance vector routing.

•In path vector routing, we assume that there is one node in each autonomous system that acts on behalf of the entire autonomous system. Let us call it the speaker node.

•The speaker node in an AS creates a routing table and advertises it to speaker nodes in the neighboring ASs.

BGP

•Border Gateway Protocol (BGP) is an interdomain routing protocol using path vector routing.



Destinat.	Path for A1	Path for B1	Path for C1	Path for D1
A1 .... A5	AS1 ... AS1	AS2 - AS1	AS3 - AS1	AS4 - AS3 - AS1
B1 .... B4	AS1 - ... AS2	AS2 - AS3	AS3 - AS2	AS4 - AS3 - AS2
C1 .... C3	AS1 ... AS3	AS2 - AS3 - ~~AS4~~	AS3 - AS3	AS4 - AS3
D1 .... D4	AS1 - AS3 - ~~AS4~~	AS2 - AS3 - AS4	AS3 - AS4	AS4 - AS4

Ques: Discuss each command in detail used in networking: **10M 2021-22**
a- ipconfig
b- netstat
c- ping
d- hostname
e- tracert

a) ipconfig:
The "ipconfig" command is used in Windows operating systems to display the IP configuration information of a network interface. It provides information about the IP address, subnet mask, default gateway, and other network-related details of the computer.

- "ipconfig /all": Displays detailed information about all network interfaces, including the physical addresses
- "ipconfig /release": Releases the current IP address, making it available for reassignment.
- "ipconfig /renew": Requests a new IP address from a DHCP server.

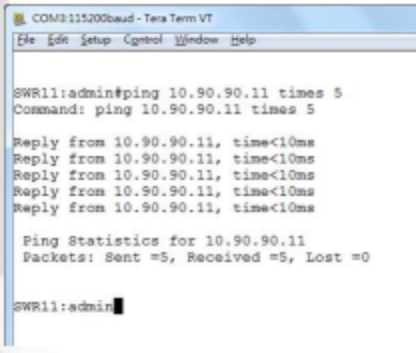
b) netstat:
The "netstat" command is a versatile networking tool available in various operating systems, including Windows and Unix-like systems. It provides information about network connections, routing tables, and network statistics.

- "netstat -a": Displays all active network connections
- "netstat -r": Shows the routing table
- "netstat -n": Displays network connections with numerical IP addresses and port numbers

c)ping:

This command is used to test the connectivity between network devices .

When you send a ping command to a specific IP address or hostname, it sends ICMP Echo Request packets to the destination and waits for ICMP Echo Reply packets.



- "ping <IP address or hostname>": Sends ICMP Echo Request packets to the specifiedIP address or hostname and displays the round-trip time (RTT) and success/failure status for each reply.

d) hostname:

The "hostname" command is used to display or set the hostname of a computer or device.

- "hostname": Displays the current hostname of the computer.
- "hostname -I": Displays the IP addresses assigned to the computer, which can be useful for identifying the IP addresses associated with the hostname.

e) tracert:

The "tracert" (short for trace route) command is used to trace the path that packets take from the local computer to a specified destination IP address or hostname.

- "tracert <IP address or hostname>": Traces the route taken by packets from the local computer to the specified destination, displaying the IP addresses or hostnames of intermediate routers along the way.
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